

Objects in JavaScript



You said:

object and state explain all concept from basics to advanced

ChatGPT said:

Objects in JavaScript (Basics → Advanced)

Objects are one of the most important concepts in JavaScript and React.

An object stores data in **key** : **value** pairs.

```
JavaScriptconst person = {  
  name: "Aamir",  
  age: 22,  
  city: "Delhi"  
};
```

Here:

- **name, age, city** → keys (properties)
- **"Aamir", 22, "Delhi"** → values

1. Why Objects Are Used

Without objects:

```
JavaScriptconst name = "Aamir";  
const age = 22;  
const city = "Delhi";
```

Hard to manage.

With object:

```
JavaScriptconst user = {  
  name: "Aamir",  
  age: 22,  
  city: "Delhi"  
};
```

Everything stays together.

2. Real Life Example

A mobile phone has:

- brand
- model
- price
- camera

```
JavaScriptconst phone = {  
  brand: "Samsung",  
  model: "S24",  
  price: 70000,  
  camera: "50MP"  
};
```

3. Accessing Object Values

Dot Notation

```
JavaScriptconsole.log(phone.brand);
```

Output:

```
JavaScriptSamsung
```

Bracket Notation

```
JavaScriptconsole.log(phone["price"]);
```

Output:

```
JavaScript70000
```

4. Updating Object Values

```
JavaScriptphone.price = 80000;  
  
console.log(phone.price);
```

Output:

```
JavaScript80000
```

5. Adding New Property

```
JavaScriptphone.color = "Black";  
  
console.log(phone);
```

6. Deleting Property

```
JavaScriptdelete phone.camera;  
  
console.log(phone);
```

7. Nested Objects

Object inside object.

```
JavaScriptconst student = {  
  name: "Aamir",  
  marks: {  
    math: 90,  
    english: 85  
  }  
};
```

Access:

```
JavaScriptconsole.log(student.marks.math);
```

Output:

```
JavaScript90
```

8. Array of Objects

Very important in React.

```
JavaScriptconst users = [  
  { name: "Aamir", age: 22 },  
  { name: "Ali", age: 25 },  
  { name: "Sara", age: 20 }  
];
```

Access:

```
JavaScriptconsole.log(users[1].name);
```

Output:

```
JavaScriptAli
```

9. Object Methods

Functions inside objects.

```
JavaScriptconst person = {  
  name: "Aamir",  
  greet: function () {  
    console.log("Hello");  
  }  
};  
  
person.greet();
```

Output:

```
JavaScriptHello
```

10. this Keyword

this means current object.

```
JavaScriptconst user = {  
  name: "Aamir",  
  sayName() {  
    console.log(this.name);  
  }  
};  
  
user.sayName();
```

Output:

```
JavaScriptAamir
```

11. Object Destructuring

Used heavily in React.

```
JavaScriptconst user = {  
  name: "Aamir",  
  age: 22  
};  
  
const { name, age } = user;  
  
console.log(name);
```

12. Spread Operator in Object

Copy object.

```
JavaScriptconst user1 = {  
  name: "Aamir",  
  age: 22  
};  
  
const user2 = {  
  ...user1,  
  city: "Delhi"  
};  
  
console.log(user2);
```

Output:

```
JavaScript{  
  name: "Aamir",  
  age: 22,  
  city: "Delhi"  
}
```

13. Object.keys()

Returns keys.

```
JavaScriptconst user = {  
  name: "Aamir",  
  age: 22  
};  
  
console.log(Object.keys(user));
```

Output:

```
JavaScript["name", "age"]
```

14. Object.values()

```
JavaScriptconsole.log(Object.values(user));
```

Output:

```
JavaScript["Aamir", 22]
```

15. Object.entries()

```
JavaScriptconsole.log(Object.entries(user));
```

Output:

```
JavaScript[["name", "Aamir"], ["age", 22]]
```

16. Loop Through Object

```
JavaScriptconst user = {  
  name: "Aamir",  
  age: 22  
};  
  
for (let key in user) {  
  console.log(key, user[key]);  
}
```

Output:

```
JavaScriptname Aamir  
age 22
```

17. Shallow Copy vs Deep Copy

Shallow Copy

```
JavaScriptconst user1 = {  
  name: "Aamir"  
};  
  
const user2 = user1;  
  
user2.name = "Ali";  
  
console.log(user1.name);
```

Output:

```
JavaScriptAli
```

Because both point to same memory.

Deep Copy

```
JavaScriptconst user1 = {  
  name: "Aamir"  
};  
  
const user2 = { ...user1 };  
  
user2.name = "Ali";  
  
console.log(user1.name);
```

Output:

```
JavaScriptAamir
```

18. Optional Chaining

Avoid errors.

```
JavaScriptconst user = {  
  address: {  
    city: "Delhi"  
  }  
};  
  
console.log(user.address?.city);
```

19. Object.freeze()

Cannot modify object.

```
JavaScriptconst user = {  
  name: "Aamir"  
};  
  
Object.freeze(user);  
  
user.name = "Ali";  
  
console.log(user.name);
```

Output:

```
JavaScriptAamir
```

20. Constructor Function

Old way of creating objects.

```
JavaScriptfunction User(name, age) {  
  this.name = name;  
  this.age = age;  
}  
  
const u1 = new User("Aamir", 22);  
  
console.log(u1);
```

21. Classes (Modern OOP)

```
JavaScriptclass User {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }

  greet() {
    console.log("Hello");
  }
}

const u1 = new User("Aamir", 22);

u1.greet();
```

STATE IN REACT (Basics → Advanced)

State is the heart of React.

State stores changing data.

Examples:

- counter value
- input text
- dark/light mode
- like button
- cart items

1. What is State?

State = data that changes in UI.

Example:

```
JavaScriptlet count = 0;
```

But React UI will NOT update automatically.

So React gives State.

2. useState Hook

```
JavaScriptimport { useState } from "react";
```

3. Creating State

```
JavaScriptconst [count, setCount] = useState(0);
```

Explanation:

Part	Meaning
count	current value
setCount	function to update value

Part	Meaning
------	---------

useState(0)	initial value
-------------	---------------

4. Counter Example

```
JavaScriptimport { useState } from "react";

function Counter() {
  const [count, setCount] = useState(0);

  return (
    <div>
      <h1>{count}</h1>

      <button onClick={() => setCount(count + 1)}>
        Increase
      </button>
    </div>
  );
}

export default Counter;
```

5. How State Works

When state changes:

1. React re-renders component
2. UI updates automatically

6. Why We Never Change State Directly

Wrong:

```
JavaScriptcount = count + 1;
```

OR

```
JavaScriptstate.name = "Ali";
```

React will not track properly.

Correct:

```
JavaScriptsetCount(count + 1);
```

7. State is Asynchronous

```
JavaScriptsetCount(count + 1);

console.log(count);
```

Old value may print.

Because React updates later.

8. Functional Update

Best practice when new state depends on old state.

```
JavaScriptsetCount(prev => prev + 1);
```

9. Multiple Updates Problem

Wrong:

```
JavaScriptsetCount(count + 1);  
setCount(count + 1);
```

Only increases once sometimes.

Correct:

```
JavaScriptsetCount(prev => prev + 1);  
setCount(prev => prev + 1);
```

Now increases twice.

10. State with String

```
JavaScriptconst [name, setName] = useState("");
```

11. State with Boolean

```
JavaScriptconst [dark, setDark] = useState(false);
```

Toggle:

```
JavaScriptsetDark(!dark);
```

12. State with Object

Very important.

```
JavaScriptconst [user, setUser] = useState({  
  name: "Aamir",  
  age: 22  
});
```

Update object:

```
JavaScriptsetUser({  
  ...user,  
  age: 23  
});
```

Why Spread Operator?

Because state should be immutable.

Wrong:

```
JavaScriptuser.age = 23;
```

Correct:

```
JavaScriptsetUser({  
  ...user,  
  age: 23  
});
```

13. State with Array

```
JavaScriptconst [items, setItems] = useState([]);
```

Add item:

```
JavaScriptsetItems([...items, "Apple"]);
```

14. Removing Item

```
JavaScriptsetItems(items.filter(item => item !== "Apple"));
```

15. Updating Array Object

```
JavaScriptconst [users, setUsers] = useState([
  { id: 1, name: "Aamir" },
  { id: 2, name: "Ali" }
]);

setUsers(
  users.map(user =>
    user.id === 1
      ? { ...user, name: "Updated" }
      : user
  )
);
```

16. Lifting State Up

Share state between components.

Parent keeps state and passes via props.

17. Controlled Components

Input controlled by state.

```
JavaScriptconst [text, setText] = useState("");

<input
  value={text}
  onChange={(e) => setText(e.target.value)}
/>
```

18. Derived State

Computed from existing state.

```
JavaScriptconst total = price * quantity;
```

Do not store unnecessarily.

19. State vs Props

State	Props
Mutable	Immutable
Inside component	Passed from parent

State	Props
Can change	Read only

20. Local State vs Global State

Local State

Inside one component.

```
JavaScriptuseState()
```

Global State

Shared across app.

Tools:

- Context API
- Redux
- Zustand

21. Batching in React

React groups state updates for performance.

```
JavaScriptsetCount(c => c + 1);  
setCount(c => c + 1);
```

React batches them.

22. State Immutability

Never mutate state directly.

React depends on reference changes.

Wrong:

```
JavaScriptuser.name = "Ali";
```

Correct:

```
JavaScriptsetUser({  
  ...user,  
  name: "Ali"  
});
```

23. React Re-render

Component re-renders when:

- state changes
- props change

- context updates

24. Common Interview Questions

Q1: Why useState returns array?

```
JavaScriptconst [state, setState] = useState();
```

Easy custom naming.

Q2: Why state updates are async?

For performance and batching.

Q3: Why immutability important?

React compares old vs new references.

Q4: Difference between state and normal variable?

Normal variable does not re-render UI.

State does.

25. Real Project Example

Like Button

```
JavaScriptfunction LikeButton() {  
  const [liked, setLiked] = useState(false);  
  
  return (  
    <button onClick={() => setLiked(!liked)}>  
      {liked ? "Liked" : "Like"}  
    </button>  
  );  
}
```

26. Advanced State Concepts

useReducer

For complex state.

```
JavaScriptconst [state, dispatch] = useReducer(reducer, initialState);
```

Context API

Avoid prop drilling.

Redux

Global state management.

27. Most Important Thing to Remember

React state should be:

- immutable
- updated with setter function

- never modified directly

Quick Revision

Object

- stores data in key-value pair
- can contain array, function, object
- used everywhere in JS & React

State

- stores changing UI data
- useState creates state
- changing state re-renders UI
- never update directly
- use spread operator for objects/arrays

You said:

ChatGPT said:

Object vs State (Easy Explanation)

Feature	Object	State
Used In	JavaScriptReact	
Purpose	Store data	Store changing UI data
Re-render UI?	No	Yes
Can Change?	Yes	Yes using setter
Syntax	{ }	useState()

1. Object in JavaScript

Object stores data in key-value pair.

```
JavaScriptconst user = {
  name: "Aamir",
  age: 22
};
```

Access values:

```
JavaScriptconsole.log(user.name);
console.log(user.age);
```

Output:

```
JavaScriptAamir
22
```

Update object:

```
JavaScriptuser.age = 23;  
  
console.log(user);
```

2. State in React

State stores changing data in UI.

```
JavaScriptimport { useState } from "react";  
  
function App() {  
  const [count, setCount] = useState(0);  
  
  return (  
    <button onClick={() => setCount(count + 1)}>  
      {count}  
    </button>  
  );  
}
```

3. Main Difference

Object

Changing object does NOT update UI automatically.

```
JavaScriptconst user = {  
  name: "Aamir"  
};  
  
user.name = "Ali";
```

UI will not know.

State

Changing state updates UI automatically.

```
JavaScriptsetCount(count + 1);
```

React re-renders component.

4. Object Inside State

Very important in React.

```
JavaScriptconst [user, setUser] = useState({  
  name: "Aamir",  
  age: 22  
});
```

Update correctly:

```
JavaScriptsetUser({  
  ...user,  
  age: 23  
});
```

5. Why Spread Operator?

Wrong:

```
JavaScriptuser.age = 23;
```

Direct mutation.

Correct:

```
JavaScriptsetUser({  
  ...user,  
  age: 23  
});
```

Creates new object.

React detects change properly.

6. Object Example

```
JavaScriptconst car = {  
  brand: "BMW",  
  color: "Black"  
};  
  
console.log(car.brand);
```

7. State Example

```
JavaScriptconst [dark, setDark] = useState(false);  
  
<button onClick={() => setDark(!dark)}>  
  Toggle  
</button>
```

8. Memory Trick

Object

Just stores data.

State

Stores data + updates screen.

9. Interview Question

Why not modify state directly?

Wrong:

```
JavaScriptuser.name = "Ali";
```

Correct:

```
JavaScriptsetUser({  
  ...user,  
  name: "Ali"  
});
```

Because React checks new reference.

10. Simple Real Life Analogy

Object

Notebook storing information.

State

Electronic display board.

When data changes, screen updates automatically.

You said:

PS F:\Apna College Delta Batch\L68ReactPart4\ClassProject\src> npm run dev > classproject@0.0.0 dev > vite 'vite' is not recognized as an internal or external command, operable program or batch file.

ChatGPT said:

This error means **Vite is not installed** in your project or `node_modules` folder is missing.

Follow these steps carefully.

Step 1: Install Dependencies

Go to your project folder:

```
Bashcd "F:\Apna College Delta Batch\L68ReactPart4\ClassProject"
```

Now run:

```
Bashnpm install
```

This installs all packages including Vite.

Step 2: Run Project

After installation:

```
Bashnpm run dev
```

Why This Error Happens

```
Bash'vite' is not recognized
```

means:

- `node_modules` folder missing
- OR Vite package not installed

If Still Not Working

Install Vite manually:

```
Bashnpm install vite --save-dev
```

Then run:

```
Bashnpm run dev
```

Check package.json

Your `package.json` should contain:

```
JSON"scripts": {
  "dev": "vite"
}
```

and

```
JSON"devDependencies": {
  "vite": "latest-version"
}
```

Most Common Fix

Usually only this works:

```
Bashnpm install
npm run dev
```

Expected Output

After success:

```
BashVITE vX.X.X ready in xxx ms

→ Local: http://localhost:5173/
```

You said:

```
import { useState } from "react"; import { v4 as uuidv4 } from "uuid"; export default function TodoList() { let [todos, setTodos] =
useState([task: "sample Task", id: uuidv4(), isDone: false]); let [newTodo, setNewTodo] = useState(""); let addNewTask = () => {
setTodos((prevTodos) => { return [...prevTodos, {task: newTodo, id: uuidv4(), isDone: false}]; }) console.log("we have to add new
Task at Todo "); setTodos([...todos, {task: newTodo, id: uuidv4()}], [...newTodo]); setNewTodo(""); }); let updateTodoValue = (event)
=> { // console.log(event.target.value); setNewTodo(event.target.value); } let deleteTodo = (id) => { console.log("we have to delete
Task at Todo ", id); setTodos(todos.filter((todo) => todo.id !== id)); } let markAsCompleted = (id) => { console.log("we have to mark
Task as completed at Todo ", id); setTodos(todos.map((todo) => todo.id === id ? {...todo, completed: true} : todo)); // let
upperCaseAll = () => { // let newArr = todos.map((todo) => { // return { // ...todo, // task: todo.task.toUpperCase(), // }; // }); //
console.log(newArr); // setTodos(newArr); // } let upperCaseAll = () => { setTodos((prevTodos) => { return prevTodos.map((todo)
=> { return { ...todo, task: todo.task.toUpperCase(), }; }); }); let markAsDone = (id) => { setTodos((todos) => todos.map((todo) =>
{ if(todo.id === id){ return { ...todo, isDone: true, }; } else { return todo; } } )); return ( <div> <h1>Todo List</h1> <input type="text"
placeholder="add a task" value={newTodo} onChange={updateTodoValue} /> <br /> <br /> <button onClick={addNewTask} >Add
Task</button> <br /> <br /> <br /> <h4>Task Todo</h4> <ul> {todos.map((todo) => ( <li key={todo.id}> <span>{todo.task}
</span> &nbsp; &nbsp; &nbsp; &nbsp; &nbsp; <button onClick={() => deleteTodo(todo.id)}>Delete</button> &nbsp; &nbsp; &nbsp; &nbsp;
&nbsp; <br /> <br /> <button onClick={() => markAsCompleted(todo.id)}>Mark As Completed</button> &nbsp; &nbsp; &nbsp; &nbsp;
&nbsp; <br /> <br /> <button onClick={() => upperCaseAll(todo.id)}>Upper Case All</button> <button onClick={() =>
markAsDone(todo.id)}>Mark As Done</button> </li> ))} </ul> <br /> <br /> <button onClick=
{upperCaseAll}>upperCaseAll</button> </div> ); } explain and write clean code
```

ChatGPT said:

Your code is good for learning React state, arrays, objects, `map()`, `filter()`, and immutability.

But there are some mistakes and unnecessary code.

Problems in Your Code

1. `setTodos()` called twice

You wrote:

```
JavaScriptsetTodos((prevTodos) => {  
  return [...prevTodos, {task: newTodo, id: uuidv4(), isDone: false}];  
})  
  
setTodos([...todos, {task: newTodo, id: uuidv4()}], [...newTodo]);
```

Second `setTodos()` is wrong and unnecessary.

Use only ONE.

2. **completed** vs **isDone**

You used:

```
JavaScriptcompleted: true
```

But initial state uses:

```
JavaScriptisDone: false
```

Property name should stay same.

Correct:

```
JavaScriptisDone: true
```

3. **upperCaseAll(todo.id)**

Function does not need `id`.

Wrong:

```
JavaScript<button onClick={() => upperCaseAll(todo.id)}>
```

Correct:

```
JavaScript<button onClick={upperCaseAll}>
```

CLEAN VERSION OF YOUR CODE

```

JavaScriptimport { useState } from "react";
import { v4 as uuidv4 } from "uuid";

export default function TodoList() {

  // State for all todos
  const [todos, setTodos] = useState([
    {
      task: "Sample Task",
      id: uuidv4(),
      isDone: false,
    },
  ]);

  // State for input field
  const [newTodo, setNewTodo] = useState("");

  // Add New Task
  const addNewTask = () => {

    // Prevent empty task
    if (newTodo.trim() === "") return;

    const newTask = {
      task: newTodo,
      id: uuidv4(),
      isDone: false,
    };

    setTodos((prevTodos) => [...prevTodos, newTask]);

    // Clear input
    setNewTodo("");
  };

  // Update input value
  const updateTodoValue = (event) => {
    setNewTodo(event.target.value);
  };

  // Delete Todo
  const deleteTodo = (id) => {
    setTodos((prevTodos) =>
      prevTodos.filter((todo) => todo.id !== id)
    );
  };

  // Mark one task as done
  const markAsDone = (id) => {
    setTodos((prevTodos) =>
      prevTodos.map((todo) =>
        todo.id === id
          ? { ...todo, isDone: true }
          : todo
      )
    );
  };

  // Convert all tasks to uppercase
  const upperCaseAll = () => {
    setTodos((prevTodos) =>
      prevTodos.map((todo) => ({
        ...todo,
        task: todo.task.toUpperCase(),
      }))
    );
  };

  return (
    <div>
      <h1>Todo List</h1>

      <input

```

[illegible]

Full Explanation

1. useState

```
JavaScriptconst [todos, setTodos] = useState([]);
```

- **todos** → current state

- `setTodos` → function to update state

2. Array of Objects

```
JavaScript[
  {
    task: "Sample Task",
    id: "123",
    isDone: false
  }
]
```

Each todo is an object.

3. Controlled Input

```
JavaScript<input
  value={newTodo}
  onChange={updateTodoValue}
/>
```

React controls input value using state.

4. Adding Todo

```
JavaScriptsetTodos((prevTodos) => [...prevTodos, newTask]);
```

Spread Operator

```
JavaScript...prevTodos
```

copies old array.

Then new task added.

5. Why Functional Update?

```
JavaScriptsetTodos((prevTodos) => ...)
```

Safest way because React state updates are asynchronous.

6. filter()

Used for deleting.

```
JavaScriptfilter((todo) => todo.id !== id)
```

Keeps all except clicked one.

7. map()

Used for updating.

```
JavaScriptmap((todo) => ...)
```

Creates new array.

8. Immutability

Never modify state directly.

Wrong:

```
JavaScripttodos.push(newTask);
```

Correct:

```
JavaScriptsetTodos([...todos, newTask]);
```

9. Conditional Styling

```
JavaScripttextDecoration: todo.isDone  
  ? "line-through"  
  : "none"
```

If task completed → line-through.

10. UUID

```
JavaScriptuuidv4()
```

Creates unique id.

Needed for:

```
JavaScriptkey={todo.id}
```

React needs unique key for list rendering.

React Concepts Used

You used:

- useState
- controlled component
- arrays in state
- objects in state
- map()
- filter()
- spread operator
- conditional rendering
- event handling
- immutable updates
- unique keys

These are core React interview concepts.

You said:

You said:

ChatGPT said:

Activity: Create Todo App in React

This activity will help you practice:

- useState
- array state
- object state
- map()
- filter()
- event handling
- controlled input
- spread operator

Goal

Build a Todo App with:

- Add Task
- Delete Task
- Mark as Done

Step 1: Create Component

Create file:

```
Todo.jsx
```

Step 2: Basic Setup

```
JavaScriptimport { useState } from "react";

export default function Todo() {

  return (
    <div>
      <h1>Todo App</h1>
    </div>
  );
}
```

Step 3: Create State

```
JavaScriptconst [todos, setTodos] = useState([]);
const [newTodo, setNewTodo] = useState("");
```

Explanation:

State	Purpose
todos	stores all tasks

State	Purpose
-------	---------

newTodo	stores input value
---------	--------------------

Step 4: Create Input Field

```
JavaScript<input
  type="text"
  placeholder="Enter task"
  value={newTodo}
  onChange={(e) => setNewTodo(e.target.value)}
/>
```

This is called:

Controlled Component

React controls input using state.

Step 5: Add Button

```
JavaScript<button onClick={addTodo}>
  Add Task
</button>
```

Step 6: Create addTodo Function

```
JavaScriptconst addTodo = () => {

  if (newTodo.trim() === "") return;

  const newTask = {
    id: Date.now(),
    task: newTodo,
    isDone: false
  };

  setTodos((prevTodos) => [
    ...prevTodos,
    newTask
  ]);

  setNewTodo("");
};
```

Important Concepts

Spread Operator

```
JavaScript...prevTodos
```

Copies old array.

Why Functional Update?

```
JavaScriptsetTodos((prevTodos) => ...)
```

Because state updates are asynchronous.

Step 7: Display Todos

```
JavaScript<ul>
  {todos.map((todo) => (
    <li key={todo.id}>
      {todo.task}
    </li>
  ))}
</ul>
```

Why map()?

`map()` converts array into UI elements.

Step 8: Delete Todo

Button

```
JavaScript<button onClick={() => deleteTodo(todo.id)}>
  Delete
</button>
```

Function

```
JavaScriptconst deleteTodo = (id) => {
  setTodos(
    todos.filter((todo) => todo.id !== id)
  );
};
```

Why filter()?

Keeps all items except deleted one.

Step 9: Mark As Done

Button

```
JavaScript<button onClick={() => markAsDone(todo.id)}>
  Done
</button>
```

Function

```
JavaScriptconst markAsDone = (id) => {

  setTodos(
    todos.map((todo) =>
      todo.id === id
        ? { ...todo, isDone: true }
        : todo
    )
  );
};
```

Why map() Here?

We update only one object.

Step 10: Add Line Through Style

```
JavaScript<span
  style={{
    textDecoration:
      todo.isDone
        ? "line-through"
        : "none"
  }}
>
  {todo.task}
</span>
```

Final Complete Code

```

JavaScriptimport { useState } from "react";

export default function Todo() {

  const [todos, setTodos] = useState([]);
  const [newTodo, setNewTodo] = useState("");

  // Add Todo
  const addTodo = () => {

    if (newTodo.trim() === "") return;

    const newTask = {
      id: Date.now(),
      task: newTodo,
      isDone: false
    };

    setTodos((prevTodos) => [
      ...prevTodos,
      newTask
    ]);

    setNewTodo("");
  };

  // Delete Todo
  const deleteTodo = (id) => {

    setTodos(
      todos.filter((todo) => todo.id !== id)
    );
  };

  // Mark As Done
  const markAsDone = (id) => {

    setTodos(
      todos.map((todo) =>
        todo.id === id
          ? { ...todo, isDone: true }
          : todo
      )
    );
  };

  return (
    <div>

      <h1>Todo App</h1>

      <input
        type="text"
        placeholder="Enter task"
        value={newTodo}
        onChange={(e) =>
          setNewTodo(e.target.value)
        }
      />

      <button onClick={addTodo}>
        Add
      </button>

      <hr />

      <ul>

        {todos.map((todo) => (

          <li key={todo.id}>

            <span

```

Concepts Used

Bonus Activities

1. Add Edit Button

2. Add Uppercase All

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3. Add Completed Count

Example:

```
Completed: 2
```

4. Add Clear All Button

Remove all todos.

```
JavaScriptsetTodos([]);
```

You said:

unique key for list items

ChatGPT said:

Unique Key for List Items in React

When using `map()` in React, every item should have a unique `key`.

Example

```
JavaScriptconst fruits = ["Apple", "Mango", "Banana"];

function App() {
  return (
    <ul>
      {fruits.map((fruit, index) => (
        <li key={index}>
          {fruit}
        </li>
      ))}
    </ul>
  );
}
```

What is key?

`key` is a special React prop.

It helps React identify each list item uniquely.

Why React Needs key?

Suppose list changes:

Before:

```
Apple
Mango
Banana
```

After deleting Mango:

```
Apple
Banana
```

React compares old vs new list.

`key` helps React know:

- which item changed

- which item deleted
- which item added

This improves:

- performance
- re-rendering
- DOM updates

Correct Way

Best:

```
JavaScript<li key={todo.id}>
```

Use unique database id or UUID.

Example with Object

```
JavaScriptconst users = [
  { id: 1, name: "Aamir" },
  { id: 2, name: "Ali" }
];

<ul>
  {users.map((user) => (
    <li key={user.id}>
      {user.name}
    </li>
  ))}
</ul>
```

Why Not Use index?

Possible:

```
JavaScriptkey={index}
```

But not best for dynamic lists.

Problem occurs when:

- deleting items
- reordering
- inserting items

React may update wrong item.

Bad Example

```
JavaScript{todos.map((todo, index) => (
  <li key={index}>
    {todo.task}
  </li>
))}
```

If first item deleted:

```
0 -> Apple
1 -> Mango
```

After delete Apple:

```
0 -> Mango
```

React gets confused.

Best Practice

Use:

- database id
- uuid
- unique identifier

Example:

```
JavaScriptkey={todo.id}
```

UUID Example

```
JavaScriptimport { v4 as uuidv4 } from "uuid";

const todo = {
  id: uuidv4(),
  task: "Learn React"
};
```

Important Interview Question

Q: Why key important in React?

Answer:

Keys help React efficiently identify, update, add, and remove list items during re-rendering.

React Warning Without Key

You may see:

```
Each child in a list should have a unique "key" prop.
```

Means React needs unique key.

Quick Revision

Key	Purpose
unique	identify item
improves performance	faster DOM updates
used in map()	list rendering
best choice	unique id

Key	Purpose
avoid	index for dynamic list

Most Common Pattern

```
JavaScript{
  todos.map((todo) => (
    <li key={todo.id}>
      {todo.task}
    </li>
  ))
}
```

You said:

ChatGPT said:

Deleting from Array in JavaScript & React

In React, deleting items from array is mostly done using:

```
JavaScriptfilter()
```

Because React state should be immutable.

1. Delete from Normal JavaScript Array

```
JavaScriptconst fruits = ["Apple", "Mango", "Banana"];
```

Using filter()

```
JavaScriptconst newFruits = fruits.filter(
  (fruit) => fruit !== "Mango"
);

console.log(newFruits);
```

Output:

```
JavaScript["Apple", "Banana"]
```

How filter() Works

filter() keeps only matching items.

```
JavaScriptfruit !== "Mango"
```

Means:

Keep everything except Mango.

Delete Item

```
JavaScriptconst deleteTodo = (task) => {  
  setTodos(  
    todos.filter((todo) => todo !== task)  
  );  
};
```

[illegible]

3. Delete Object from Array

Very important.

Array of Objects

```
JavaScriptconst users = [  
  { id: 1, name: "Aamir" },  
  { id: 2, name: "Ali" }  
];
```

Delete by id

```
JavaScriptsetUsers(  
  users.filter((user) => user.id !== id)  
);
```

Example

```
JavaScriptconst deleteUser = (id) => {  
  
  setUsers(  
    users.filter((user) => user.id !== id)  
  );  
  
};
```

Why Use id?

Because ids are unique.

Names may repeat.

4. Why Not Use splice()?

Wrong in React:

```
JavaScripttodos.splice(index, 1);
```

Because:

- modifies original array
- React may not re-render properly

React Rule

✗ Never mutate state directly

Wrong:

```
JavaScripttodos.pop();  
todos.splice();
```

Correct

✓ Create new array

```
JavaScriptfilter()  
map()  
spread (...)
```

5. Functional Update (Best Practice)

```
JavaScriptsetTodos((prevTodos) =>
  prevTodos.filter(
    (todo) => todo.id !== id
  )
);
```

Why Better?

Because state updates are asynchronous.

Safer with previous state.

6. Visual Flow

```
Old Array
↓
filter removes item
↓
New Array created
↓
setState(newArray)
↓
React re-renders UI
```

7. Real Todo Example

```
JavaScriptconst deleteTodo = (id) => {

  setTodos((prevTodos) =>
    prevTodos.filter(
      (todo) => todo.id !== id
    )
  );
};
```

8. Common Interview Question

Q: Why use filter() for deleting in React?

Answer:

Because filter creates a new array without mutating original state.

Quick Revision

Method	Use
filter()	delete
map()	update
spread (...)	copy/add
splice()	avoid in React

Most Important Pattern

```
JavaScriptsetState(
  prev =>
    prev.filter(item => condition)
);
```

You said:

ChatGPT said:

Update All Items in Array (React)

To update all items in an array, we mostly use:

```
JavaScriptmap()
```

Because `map()` creates a new updated array.

1. Normal JavaScript Example

```
JavaScriptconst nums = [1, 2, 3];
```

Update all values:

```
JavaScriptconst updatedNums = nums.map((num) => {  
  return num * 2;  
});  
  
console.log(updatedNums);
```

Output:

```
JavaScript[2, 4, 6]
```

2. Update All Items in React State

```
JavaScriptconst [fruits, setFruits] = useState([  
  "apple",  
  "mango",  
  "banana"  
]);
```

Convert All to Uppercase

```
JavaScriptsetFruits(  
  fruits.map((fruit) =>  
    fruit.toUpperCase()  
  )  
);
```

Full Example

```
JavaScriptimport { useState } from "react";

export default function App() {

  const [fruits, setFruits] = useState([
    "apple",
    "mango",
    "banana"
  ]);

  const upperCaseAll = () => {

    setFruits(
      fruits.map((fruit) =>
        fruit.toUpperCase()
      )
    );

  };

  return (
    <div>

      <button onClick={upperCaseAll}>
        UpperCase All
      </button>

      <ul>
        {fruits.map((fruit, index) => (
          <li key={index}>
            {fruit}
          </li>
        ))}
      </ul>

    </div>
  );
}
```

3. Update All Objects in Array

Very important in React.

Example Array

```
JavaScriptconst users = [
  { id: 1, name: "Aamir" },
  { id: 2, name: "Ali" }
];
```

Update All Names to Uppercase

```
JavaScriptsetUsers(
  users.map((user) => ({
    ...user,
    name: user.name.toUpperCase()
  }))
);
```

How It Works

map()

Loops through every item.

Spread Operator

```
JavaScript...user
```

Copies old object.

Update Property

```
JavaScriptname: user.name.toUpperCase()
```

Changes only name.

4. Todo Example

```
JavaScriptconst upperCaseAll = () => {  
  
  setTodos((prevTodos) => {  
  
    return prevTodos.map((todo) => {  
  
      return {  
        ...todo,  
        task: todo.task.toUpperCase(),  
      };  
    });  
  });  
};
```

Before

```
JavaScript[  
  { task: "study" },  
  { task: "coding" }  
]
```

After

```
JavaScript[  
  { task: "STUDY" },  
  { task: "CODING" }  
]
```

5. Why Not Modify Directly?

Wrong:

```
JavaScripttodos[0].task = "NEW";
```

Because React state should be immutable.

Correct

```
JavaScriptmap()  
spread (...)  
setState()
```

6. Functional Update (Best Practice)

```
JavaScriptsetTodos((prevTodos) =>
  prevTodos.map((todo) => ({
    ...todo,
    task: todo.task.toUpperCase(),
  }))
);
```

Why Better?

Because state updates are asynchronous.

7. Visual Flow

```
Old Array
↓
map() updates every item
↓
New Array created
↓
setState(newArray)
↓
React re-renders UI
```

8. Common Interview Question

Q: Why use map() for updating array in React?

Answer:

Because map creates a new updated array without mutating original state.

Quick Revision

Operation	Method
Add	spread (...)
Delete	filter()
Update One	map()
Update All	map()

Most Important Pattern

```
JavaScriptsetState(
  prev =>
    prev.map(item => ({
      ...item,
      updatedValue
    }))
);
```

You said:

ChatGPT said:

Update One Item in Array (React)

To update one item inside an array, we use:

```
JavaScriptmap()
```

because `map()` creates a new updated array.

1. Basic Idea

Suppose:

```
JavaScript[
  { id: 1, name: "Aamir" },
  { id: 2, name: "Ali" }
]
```

We want to update only:

```
JavaScriptid: 2
```

2. Using map()

```
JavaScriptarray.map((item) => {
  if(condition){
    return updatedItem;
  }
  return item;
})
```

3. React Example

```
JavaScriptconst [users, setUsers] = useState([
  { id: 1, name: "Aamir" },
  { id: 2, name: "Ali" }
]);
```

Update One User

```
JavaScriptconst updateUser = (id) => {
  setUsers(
    users.map((user) =>
      user.id === id
        ? { ...user, name: "Updated" }
        : user
    )
  );
};
```

How It Works

Step 1 → map()

Loops through every item.

Step 2 → Condition

```
JavaScriptuser.id === id
```

Checks which item to update.

```
JavaScript{ ...user, name: "Updated" }
```

4. Full Example

5. Todo Example

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```
JavaScriptconst markAsDone = (id) => {

  setTodos(

    todos.map((todo) =>

      todo.id === id
      ? { ...todo, isDone: true }
      : todo

    )

  );

};
```

Before

```
JavaScript[
  { task: "Study", isDone: false },
  { task: "Coding", isDone: false }
]
```

After Updating One

```
JavaScript[
  { task: "Study", isDone: true },
  { task: "Coding", isDone: false }
]
```

Only one item updated.

6. Why Not Modify Directly?

Wrong:

```
JavaScripttodos[0].isDone = true;
```

Because React state should be immutable.

Correct Way

```
JavaScriptmap()
spread (...)
setState()
```

7. Functional Update (Best Practice)

```
JavaScriptsetTodos((prevTodos) =>

  prevTodos.map((todo) =>

    todo.id === id
    ? { ...todo, isDone: true }
    : todo

  )

);
```

Why Better?

Because state updates are asynchronous.

8. Visual Flow

```

Old Array
↓
map() loops through all items
↓
Condition finds matching item
↓
Create updated object
↓
Return new array
↓
setState(newArray)
↓
React re-renders UI

```

9. Common Interview Question

Q: Why use map() for updating one item?

Answer:

Because map creates a new updated array without mutating original state.

10. Quick Revision

Operation	Method
Add	spread (...)
Delete	filter()
Update One	map()
Update All	map()

Most Important Pattern

```

JavaScriptsetState(

  prev =>

  prev.map((item) =>

    item.id === id
    ? { ...item, updatedValue }
    : item

  )

);

```

You said:

add mark as done features

ChatGPT said:

Add “Mark As Done” Feature in React Todo App

We use:

- `map()` → update one item
- `isDone` → track completed task

- conditional styling → line-through

1. Todo State

```
JavaScriptconst [todos, setTodos] = useState([
  {
    id: 1,
    task: "Study React",
    isDone: false
  }
]);
```

Why isDone?

```
JavaScriptisDone: false
```

Means task is not completed.

Later:

```
JavaScriptisDone: true
```

Means completed.

2. Add Mark As Done Function

```
JavaScriptconst markAsDone = (id) => {
  setTodos((prevTodos) =>
    prevTodos.map((todo) =>
      todo.id === id
        ? { ...todo, isDone: true }
        : todo
    )
  );
};
```

How It Works

map()

Loops through all todos.

Condition

```
JavaScripttodo.id === id
```

Finds clicked todo.

Spread Operator

```
JavaScript{ ...todo, isDone: true }
```

Copies old object and updates **isDone**.

3. Add Button

```
JavaScript<button onClick={() => markAsDone(todo.id)}>
  Mark As Done
</button>
```

4. Add Conditional Styling

```
JavaScript<span
  style={{
    textDecoration:
      todo.isDone
        ? "line-through"
        : "none"
  }}
>
  {todo.task}
</span>
```

If isDone = true

Study React

becomes:

~~Study React~~

5. Full Working Example

```

JavaScriptimport { useState } from "react";

export default function Todo() {

  const [todos, setTodos] = useState([
    {
      id: 1,
      task: "Study React",
      isDone: false
    }
  ]);

  const [newTodo, setNewTodo] = useState("");

  // Add Todo
  const addTodo = () => {

    if (newTodo.trim() === "") return;

    const newTask = {
      id: Date.now(),
      task: newTodo,
      isDone: false
    };

    setTodos((prevTodos) => [
      ...prevTodos,
      newTask
    ]);

    setNewTodo("");
  };

  // Mark As Done
  const markAsDone = (id) => {

    setTodos((prevTodos) =>

      prevTodos.map((todo) =>

        todo.id === id
          ? { ...todo, isDone: true }
          : todo

      )

    );
  };

  return (
    <div>

      <h1>Todo App</h1>

      <input
        type="text"
        placeholder="Enter task"
        value={newTodo}
        onChange={(e) =>
          setNewTodo(e.target.value)
        }
      />

      <button onClick={addTodo}>
        Add
      </button>

      <hr />

      <ul>

        {todos.map((todo) => (

```

[illegible]

6. Important React Concepts Used

Concept	Used
useState	✓
array state	✓
object state	✓
map()	✓
spread operator	✓
event handling	✓
conditional styling	✓
immutable updates	✓

7. Most Important Pattern

```
JavaScriptsetState(  
  
  prev =>  
  
    prev.map(item =>  
  
      item.id === id  
        ? { ...item, updatedValue }  
        : item  
  
    )  
  
);
```

You said:

ChatGPT said:

Add, Delete, Update Features in React Todo App

You will learn:

- Add task
- Delete task
- Update task
- useState
- map()
- filter()

- spread operator

Full Working Todo App

```
JavaScriptimport { useState } from "react";

export default function Todo() {

  // State
  const [todos, setTodos] = useState([]);
  const [newTodo, setNewTodo] = useState("");

  // Add Task
  const addTodo = () => {

    if (newTodo.trim() === "") return;

    const newTask = {
      id: Date.now(),
      task: newTodo,
      isDone: false
    };

    setTodos((prevTodos) => [
      ...prevTodos,
      newTask
    ]);

    setNewTodo("");
  };

  // Delete Task
  const deleteTodo = (id) => {

    setTodos((prevTodos) =>
      prevTodos.filter(
        (todo) => todo.id !== id
      )
    );

  };

  // Update Task
  const updateTodo = (id) => {

    setTodos((prevTodos) =>

      prevTodos.map((todo) =>

        todo.id === id
        ? {
            ...todo,
            task: todo.task + " Updated"
          }
        : todo

      )

    );

  };

  return (
    <div>

      <h1>Todo App</h1>

      <input
        type="text"
        placeholder="Enter task"
        value={newTodo}
        onChange={(e) =>
          setNewTodo(e.target.value)
        }
      />

    </div>
  );
}
```

1. Add Feature

What Happens?

- ## 2. Delete Feature

filter()

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Before

```
JavaScript[
  { id: 1 },
  { id: 2 }
]
```

Delete id 1.

After

```
JavaScript[
  { id: 2 }
]
```

3. Update Feature

```
JavaScriptsetTodos(
  prevTodos.map((todo) =>
    todo.id === id
    ? {
      ...todo,
      task: todo.task + " Updated"
    }
    : todo
  )
);
```

map() Logic

If id matches

Update object.

Else

Return old object.

Before

```
JavaScript[
  { id: 1, task: "Study" }
]
```

After

```
JavaScript[
  { id: 1, task: "Study Updated" }
]
```

4. Why Spread Operator?

```
JavaScript...todo
```

Copies old object.

Then updates only required property.

5. Why Not Modify Directly?

Wrong:

```
JavaScripttodos[0].task = "New";
```

Because React state should be immutable.

Correct

```
JavaScriptmap()
filter()
spread (...)
```

6. Important React Concepts

Feature	Method
Add	spread (...)
Delete	filter()
Update	map()

7. Most Important Patterns

Add

```
JavaScriptsetState(prev => [
  ...prev,
  newItem
]);
```

Delete

```
JavaScriptsetState(prev =>
  prev.filter(item => condition)
);
```

Update

```
JavaScriptsetState(prev =>

  prev.map(item =>

    condition
    ? updatedItem
    : item

  )

);
```

You said:

HomeworkActivity(React)Qs.CreateaLotteryGameintheformofReactComponents.IntheUI,showabuttonthatgeneratesaLotteryTicket.Thelotteryti
lottery ticket = 248 get new ticket case 2 wining lottery congratulation you won lottery ticket = 465 get new ticket
Thesolution&approachtosolvingthiswillbecoveredintheupcomingchapter

ChatGPT said:

React Homework Activity — Lottery Game

You will practice:

- Components
- Props
- useState
- Events
- Random numbers
- Conditional rendering

Goal

Create Lottery Ticket like:

248

Check:

$2 + 4 + 8 = 14$

Not Winner.

Winning Example:

465

Because:

$4 + 6 + 5 = 15$

Winner 🎉

Complete React Code

```

JavaScriptimport { useState } from "react";

export default function LotteryGame() {

  // Generate random ticket
  const generateTicket = () => {

    return {
      n1: Math.floor(Math.random() * 10),
      n2: Math.floor(Math.random() * 10),
      n3: Math.floor(Math.random() * 10),
    };

  };

  // State
  const [ticket, setTicket] = useState(generateTicket());

  // Generate New Ticket
  const buyTicket = () => {
    setTicket(generateTicket());
  };

  // Sum of digits
  const sum =
    ticket.n1 +
    ticket.n2 +
    ticket.n3;

  // Winning condition
  const isWinner = sum === 15;

  return (
    <div>

      <h1>Lottery Game</h1>

      <h2>
        Lottery Ticket =
        {""}
        {ticket.n1}
        {ticket.n2}
        {ticket.n3}
      </h2>

      <button onClick={buyTicket}>
        Get New Ticket
      </button>

      <hr />

      {
        isWinner ? (

          <h2>
            🎉 Congratulations! You Won
          </h2>

          ) : (

            <h3>
              ❌ No Winning
            </h3>

            )
          }

    </div>
  );
}

```

How This Works

1. Random Number Generation

```
JavaScriptMath.floor(Math.random() * 10)
```

Generates:

0 to 9

2. Ticket Object

```
JavaScript{  
  n1: 4,  
  n2: 6,  
  n3: 5  
}
```

3. useState

```
JavaScriptconst [ticket, setTicket] = useState(generateTicket());
```

Stores current lottery ticket.

4. Generate New Ticket

```
JavaScriptsetTicket(generateTicket());
```

Creates new random ticket.

5. Sum of Digits

```
JavaScriptconst sum =  
  ticket.n1 +  
  ticket.n2 +  
  ticket.n3;
```

6. Winning Condition

```
JavaScriptconst isWinner = sum === 15;
```

If sum becomes 15 → Winner.

7. Conditional Rendering

```
JavaScriptisWinner ? <Winner /> : <Lose />
```

React shows different UI.

Example Outputs

Case 1 — No Winning

Lottery Ticket = 248

2 + 4 + 8 = 14

 No Winning

Case 2 — Winning

Lottery Ticket = 465

4 + 6 + 5 = 15

 Congratulations! You Won

Concepts Used

Concept	Used
useState	
components	
events	
random numbers	
conditional rendering	
object state	

Bonus Challenge

Try yourself:

1. Make 5-digit lottery

Example:

58391

2. Add Ticket Count

Tickets Generated: 5

3. Add Sound or Emoji

Winner:



Most Important Logic

JavaScriptsum === 15

This decides winner or loser.

You said:

```
import { useState } from "react"; // eslint-disable-next-line no-unused-vars import { genTicket, sum } from "./helper"; import Ticket
from "./Ticket"; // eslint-disable-next-line react/prop-types export default function Lottery({ n = 3, winCondition }) { // Initialize tickets
as an array of n tickets let [ticket, setTicket] = useState(genTicket(n)) // Check if any ticket is a winning ticket let isWinning =
winCondition(ticket); // Function to add a new ticket let buyTicket = () => { setTicket([ genTicket(n)]); }; return ( <div> <h1>Lottery
Game</h1> <Ticket ticket={ticket} /> <div className="ticket"> {ticket.map((ticket, idx) => ( <Ticket key={idx} ticket={ticket} /> ))}
</div> <button onClick={buyTicket}>Buy New Ticket</button> <h3>Current Balance: {ticket.length}</h3> <h3>{isWinning &&
"Congratulations, You won!"}</h3> </div> ); } correct and clean code
```

ChatGPT said:

Your code has some mistakes with:

- state structure
- mapping
- duplicate Ticket rendering
- buyTicket logic

You only need ONE ticket, not array of tickets.

Problems in Your Code

1. Wrong State Update

You wrote:

```
JavaScriptsetTicket([ genTicket(n)]);
```

This makes ticket an array inside array structure.

Wrong.

2. Duplicate Rendering

You already rendered:

```
JavaScript<Ticket ticket={ticket} />
```

Then again:

```
JavaScriptticket.map(...)
```

Not needed.

3. ticket.map() Error

ticket is one ticket array like:

```
JavaScript[1,2,3]
```

not array of tickets.

So:

```
JavaScriptticket.map(...)
```

is wrong here.

4. Current Balance Wrong

```
JavaScriptticket.length
```

This only shows number of digits.

Not balance.

CLEAN & CORRECT VERSION

Lottery.jsx

```
JavaScriptimport { useState } from "react";
import { genTicket } from "../helper";
import Ticket from "../Ticket";

// eslint-disable-next-line react/prop-types
export default function Lottery({
  n = 3,
  winCondition,
}) {

  // Single ticket state
  const [ticket, setTicket] = useState(
    genTicket(n)
  );

  // Check winning condition
  const isWinning = winCondition(ticket);

  // Generate new ticket
  const buyTicket = () => {
    setTicket(genTicket(n));
  };

  return (
    <div>

      <h1>Lottery Game</h1>

      <Ticket ticket={ticket} />

      <br />

      <button onClick={buyTicket}>
        Buy New Ticket
      </button>

      <br />
      <br />

      {
        isWinning && (
          <h2>
            🎉 Congratulations, You Won!
          </h2>
        )
      }

    </div>
  );
}
```

Ticket.jsx

```
JavaScriptexport default function Ticket({ ticket }) {  
  
  return (  
    <div>  
  
      <h2>  
        {ticket.join(" ")}  
      </h2>  
  
    </div>  
  );  
}
```

helper.js

```
JavaScriptexport function genTicket(n) {  
  
  let arr = [];  
  
  for (let i = 0; i < n; i++) {  
    arr.push(  
      Math.floor(Math.random() * 10)  
    );  
  }  
  
  return arr;  
}  
  
export function sum(arr) {  
  
  return arr.reduce(  
    (sum, num) => sum + num,  
    0  
  );  
}
```

App.jsx

```
JavaScriptimport Lottery from "./Lottery";  
import { sum } from "./helper";  
  
function App() {  
  
  const winCondition = (ticket) => {  
    return sum(ticket) === 15;  
  };  
  
  return (  
    <Lottery  
      n={3}  
      winCondition={winCondition}  
    />  
  );  
}  
  
export default App;
```

How This Works

Ticket Structure

```
JavaScript[4, 6, 5]
```

Winning Logic

```
JavaScript4 + 6 + 5 = 15
```

Winner 🎉

useState

```
JavaScriptconst [ticket, setTicket]
```

Stores current ticket.

Generate New Ticket

```
JavaScriptsetTicket(genTicket(n));
```

Creates new random ticket.

join()

```
JavaScriptticket.join(" ")
```

Converts:

```
JavaScript[4,6,5]
```

into:

```
4 6 5
```

Important Concepts Used

Concept	Used
useState	✓
props	✓
components	✓
conditional rendering	✓
arrays	✓
map/reduce	✓
event handling	✓

Expected Output

Losing

```
Lottery Game

2 4 8

Buy New Ticket

(No message)
```

Winning

Lottery Game

4 6 5

 Congratulations, You Won!

ChatGPT can make mistakes. Check important info.